

Q1. What is statistics and why is it important in data analytics?

Answer: Statistics is the science of collecting, organizing, analyzing, and interpreting data to make decisions. In data analytics, it provides the mathematical foundation to understand patterns, test hypotheses, and draw meaningful conclusions from data.

Q2. What is the difference between descriptive and inferential statistics?

Answer: Descriptive statistics summarizes and describes the features of a dataset using measures like mean, median, and standard deviation. Inferential statistics uses sample data to make predictions or inferences about a larger population.

Q3. What is the difference between a population and a sample in statistics?

Answer: A population is the entire group you want to draw conclusions about. A sample is a smaller subset selected from the population. Since studying the entire population is often impractical, we analyze samples and generalize the findings.

Q4. What is the difference between qualitative and quantitative data?

Answer: Qualitative data is categorical and describes characteristics or labels, such as color or gender. Quantitative data is numerical and represents measurable quantities, such as height or sales figures.

Q5. What are the four levels of measurement in statistics?

Answer: The four levels are nominal (categories with no order), ordinal (categories with order but no equal spacing), interval (ordered with equal spacing but no true zero), and ratio (ordered, equal spacing, and a true zero point).

Q6. What is the mean, median, and mode and when should each be used?

Answer: Mean is the average of all values and is used for symmetric data without outliers. Median is the middle value when data is sorted and is preferred for skewed data or when outliers exist. Mode is the most frequent value and is used for categorical data.

Q7. What is the difference between variance and standard deviation?

Answer: Variance measures the average squared deviation of each data point from the mean. Standard deviation is the square root of variance and is expressed in the same units as the data, making it more interpretable for practical use.

Q8. What is the coefficient of variation and when is it useful?

Answer: The coefficient of variation (CV) is the ratio of the standard deviation to the mean, expressed as a percentage. It is useful for comparing variability between datasets that have different units or widely different means.

Q9. What is a normal distribution and why is it so important in statistics?

Answer: A normal distribution is a symmetric, bell-shaped distribution where most values cluster around the mean. It is important because many natural phenomena follow it and many statistical tests assume normality, making it the foundation of classical statistics.

Q10. What is the empirical rule (68-95-99.7 rule) in a normal distribution?

Answer: The empirical rule states that in a normal distribution, approximately 68% of data falls within one standard deviation of the mean, 95% within two, and 99.7% within three. It is used to quickly understand the spread of normally distributed data.

Q11. What is skewness and how does it affect the relationship between mean, median, and mode?

Answer: Skewness measures the asymmetry of a distribution. In a right-skewed distribution, the mean is greater than the median, which is greater than the mode. In a left-skewed distribution, the order reverses. Skewness indicates which measure of central tendency is most appropriate.

Q12. What is kurtosis and what does it tell us about a distribution?

Answer: Kurtosis measures the heaviness of the tails of a distribution compared to a normal distribution. High kurtosis (leptokurtic) means more extreme outliers. Low kurtosis (platykurtic) means lighter tails. Normal kurtosis is called mesokurtic.

Q13. What is a box plot and what information does it communicate?

Answer: A box plot displays the five-number summary: minimum, first quartile (Q1), median, third quartile (Q3), and maximum. It visually shows the spread, center, and skewness of data and highlights potential outliers beyond the whiskers.

Q14. What is the interquartile range (IQR) and how is it used to detect outliers?

Answer: IQR is the difference between the third quartile (Q3) and first quartile (Q1), representing the middle 50% of data. Outliers are typically defined as values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$.

Q15. What is the difference between a parameter and a statistic?

Answer: A parameter is a numerical value that describes a characteristic of an entire population, such as the population mean. A statistic is a numerical value calculated from a sample, such as the sample mean, used to estimate the parameter.

Q16. What is sampling error and how can it be reduced?

Answer: Sampling error is the difference between a sample statistic and the true population parameter, arising because a sample does not perfectly represent the population. It can be reduced by increasing the sample size or using better sampling methods.

Q17. What is the Central Limit Theorem and why is it fundamental in statistics?

Answer: The Central Limit Theorem states that the distribution of sample means approaches a normal distribution as the sample size increases, regardless of the population's original distribution. It allows analysts to apply normal-distribution-based methods to a wide variety of problems.

Q18. What is the difference between simple random sampling and stratified sampling?

Answer: Simple random sampling gives every member of the population an equal chance of being selected. Stratified sampling divides the population into subgroups (strata) and randomly samples from each, ensuring all groups are represented proportionally.

Q19. What is a hypothesis test and what are the steps involved?

Answer: A hypothesis test is a procedure to evaluate whether sample data provides enough evidence to reject a claim about a population. The steps are: state the null and alternative hypotheses, choose a significance level, compute the test statistic, find the p-value, and make a decision.

Q20. What is the null hypothesis and the alternative hypothesis?

Answer: The null hypothesis (H_0) is the default assumption that there is no effect or no difference in the population. The alternative hypothesis (H_1) is the claim we want to test, suggesting there is an effect or a difference.

Q21. What is a p-value and how is it interpreted?

Answer: The p-value is the probability of obtaining results at least as extreme as the observed results, assuming the null hypothesis is true. A small p-value (typically below 0.05) suggests strong evidence against the null hypothesis, leading to its rejection.

Q22. What is the significance level (alpha) in hypothesis testing?

Answer: The significance level (alpha) is the threshold probability used to decide whether to reject the null hypothesis. A common value is 0.05, meaning there is a 5% risk of rejecting the null hypothesis when it is actually true (Type I error).

Q23. What is the difference between a one-tailed and a two-tailed test?

Answer: A one-tailed test checks for an effect in only one direction (greater than or less than). A two-tailed test checks for an effect in both directions. Two-tailed tests are more conservative and are used when the direction of the effect is not known in advance.

Q24. What is a confidence interval and how should it be interpreted?

Answer: A confidence interval is a range of values within which the true population parameter is expected to fall with a specified level of confidence. A 95% confidence interval means that if the same study were repeated 100 times, approximately 95 of those intervals would contain the true parameter.

Q25. What is Pearson's correlation coefficient and what does its value indicate?

Answer: Pearson's correlation coefficient (r) measures the strength and direction of the linear relationship between two variables. It ranges from -1 (perfect negative correlation) to +1 (perfect positive correlation), with 0 indicating no linear relationship.

Q26. What is Spearman's rank correlation and when is it used instead of Pearson's?

Answer: Spearman's rank correlation measures the strength of the monotonic relationship between two variables using their ranks instead of actual values. It is used when data is ordinal, not normally distributed, or contains significant outliers.

Q27. What is the difference between the t-test, ANOVA, and chi-square test?

Answer: A t-test compares means between two groups. ANOVA (Analysis of Variance) compares means across three or more groups. Chi-square test checks for relationships between two categorical variables. The choice depends on the number of groups and the type of data being analyzed.

Q28. What is a probability distribution and what are some common ones used in data analysis?

Answer: A probability distribution describes how the values of a random variable are distributed. Common distributions include the normal distribution (continuous, symmetric), binomial distribution (discrete, two outcomes), Poisson distribution (count of events in a fixed interval), and exponential distribution (time between events).

Q29. What is the difference between discrete and continuous probability distributions?

Answer: A discrete distribution describes variables that can only take specific countable values, such as the number of customer complaints. A continuous distribution describes variables that can take any value within a range, such as temperature or weight.

Q30. What is the difference between parametric and non-parametric statistical tests?

Answer: Parametric tests assume the data follows a specific distribution (usually normal) and work with parameters like mean and variance. Non-parametric tests make no distributional assumptions and use ranks or medians, making them suitable for ordinal data or non-normal distributions.